

3 (**short**) An inextensible cable is hung between two fixed supports 100 m apart as shown in Fig. 3. The cable is loaded with a vertical point load of 20 kN at the cable centre and with a distributed load of 1 kN m^{-1} over the right-hand half of the cable.

(a) If the dip at midspan is 1 m, calculate the support reactions. [7]

(b) Calculate the dip at a point 10 m from the right-hand support. [3]

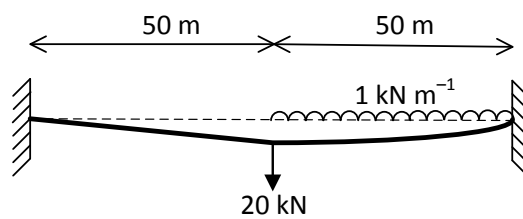


Fig. 3